

ANISH RAY

*PhD Student in Mathematics, University of Houston
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PERSONAL DETAILS

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Date of Birth May 27, 2000
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EDUCATION

Ph.D. in Mathematics **August 2025 – Present**
University of Houston, USA
Current GPA: 3.778

M2 Mathématiques et Applications, Spécialité Analyse, Arithmétique, Géométrie **2024 – 2025**
Institut de mathématique d'Orsay, Université Paris-Saclay, France
Overall grade: 12.01 (*Assez Bien*)

M.Sc. in Mathematics **2020 – 2023**
Universität Münster, Germany
Overall grade: 1.8 (*Gut*)

B.Sc. (Hons.) in Mathematics and Computing **2017 – 2020**
Institute of Mathematics and Applications, Bhubaneswar (IMA), affiliated with Utkal University, Bhubaneswar, Odisha, India
Grade: First Class with Distinction

RESEARCH INTERESTS

Analytic Number Theory, L -functions, Spectral Theory of Automorphic Forms, Automorphic Representations, Arithmetic Statistics, Equidistribution Problems, and Homogeneous Dynamics.

PREPRINTS

1. *On the Constants of the Lang-Trotter Conjecture for CM Elliptic Curves.*
arXiv:2309.09938 [math.NT]

RESEARCH AND READING PROJECTS

PIMS – First Year Interest Groups **October 2025 – May 2026**
Group Reading Project on Primes and the Theory of the Riemann Zeta Function, Pacific Institute for the Mathematical Sciences
Supervisor: Dr. Nicol Leong, Postdoctoral Fellow, University of Lethbridge

M2 Thesis **December 2024 – June 2025**
On Subconvexity of L -functions on $U(n) \times U(n+1)$ in the Depth Aspect
Supervisor: Prof. Farrell Brumley, IMJ-PRG, Sorbonne Université, France Grade: 14.5/20 (*Bien*)

Master's Thesis**September 2022 – March 2023***On the Constants of the Lang-Trotter Conjecture for CM Elliptic Curves*

Supervisor: Prof. Dr. Christopher Deninger, Mathematisches Institut, Universität Münster, Germany Grade: 1.0 (Highest Grade)

Research Project**April 2023 – April 2024***Asymptotics of the Number of Solutions of Forms Having Prime Coordinates via the Hardy-Littlewood Circle Method*

Supervisor: Prof. Julia Brandes, Chalmers University of Technology, Gothenburg, Sweden

Research Project**November 2022 – October 2023***Developed a New Large Sieve Inequality by Introducing Suitable Weight Functions*

Supervisor: Prof. Dr. Jan-Christoph Schlage-Puchta, Universität Rostock, Germany

Bachelor's Project**February 2020 – May 2020***Some Problems on Elementary Number Theory*

Supervisor: Dr. Sudhansu Sekhar Rout (currently Asst. Prof., NIT Calicut, Kerala, India)

Summer Project (Visiting Summer Research Programme)**June 2019 – July 2019***Reading Project on the Spectral Theory of Unbounded Operators*

Supervisor: Dr. Shyam Ghoshal, TIFR Centre for Applicable Mathematics (CAM), Bengaluru, India

ACADEMIC AWARDS

AMS–NSF ICM 2026 Graduate Student Travel Grant**2026***Awarded by the American Mathematical Society supporting travel for poster presentation at ICM 2026 in Philadelphia.***Presidential Fellowship****2025 – 2027***Awarded by the University of Houston to recruit doctoral students with outstanding credentials. The fellowship provides \$2,000 in addition to the Graduate Assistantship for the first two years of the PhD programme.***Graduate Tuition Fellowship****2025 – Present***Competitive fellowship awarded by the University of Houston Graduate School covering in-state tuition and mandatory fees for PhD students.***M2 Internship Allowance****2025***Awarded by the Institut de Mathématiques de Jussieu-Paris Rive Gauche (IMJ-PRG) to support thesis completion.*

Monthly allowance of €660 for three months.

IDEX Excellence Scholarship**2024 – 2025***Awarded by the Université Paris-Saclay to only 80 international students pursuing a master's degree.*

Total of €10,000 for ten months, plus €850 in travel expenses.

Graduation Scholarship**2021 – 2022***Awarded by the International Office, WWU Münster.*

Total of €1,500 for three months.

Merit Scholarship**2015***One-time merit scholarship of INR 5,000 awarded by the Kendriya Vidyalaya Sangathan (KVS) for securing 10 CGPA in AISSE 2015.***CONFERENCE AND SEMINAR TALKS**

International Congress of Mathematicians**2026***Philadelphia*

Poster presentation: “A Computational Comparison of Lang-Trotter and Hardy-Littlewood Constants for CM Elliptic Curves” at the ICM in July 2026.

Graduate Student Paper Presentation <i>Organized by the UH-AMS Graduate Chapter, University of Houston</i>	2026
Probabilistic Number Theory: Law of Large Numbers and CLT in Number Theory <i>Universität Münster</i>	2022
Linear Algebraic Groups: Connected Solvable Groups and Maximal Tori <i>Universität Münster</i>	2022
Elliptic Curves, Tate Module and the Weil Pairing <i>Universität Münster</i>	2022
Class Field Theory <i>Universität Münster</i>	2021
Zeta and L-functions in Algebraic Number Theory <i>Universität Münster</i>	2021

CONFERENCES, SUMMER SCHOOLS, AND WORKSHOPS ATTENDED

Recent Perspectives on Moments of L-functions <i>University of Texas at Dallas</i>	2026
Workshop on Automorphic Forms and Number Theory <i>IMJ-PRG, Sorbonne University, Paris</i>	2025
Summer School on Inclusive Paths in Explicit Number Theory <i>Banff International Research Station (BIRS) at the University of British Columbia – Okanagan, Canada</i>	2023
Summer School on Automorphic Forms <i>Alfréd Rényi Institute of Mathematics, Budapest</i>	2022

TEACHING

Teaching Assistant, University of Houston <i>MATH 2415 – Calculus III</i>	Fall 2026
Teaching Assistant, University of Houston <i>MATH 2451 – Accelerated Calculus II</i>	Spring 2026
Teaching Assistant, University of Houston <i>MATH 2305 – Discrete Mathematics</i>	Fall 2025, Spring 2026
Associate Faculty, Cheenta Ganit Kendra (https://cheenta.com/) <i>Online tutoring platform for aspirants of Mathematics Olympiads and national-level entrance examinations.</i> Conducted lectures, problem-solving sessions, and created and graded assignments, quizzes, and examinations.	2017 – 2018

ACADEMIC SERVICE AND OUTREACH

Proofreader, University of Houston Mathematics Contest <i>The University of Houston hosts a major annual Mathematics Contest for high school and middle school students, featuring individual and team examinations across multiple tracks.</i> https://mathcontest.uh.edu/	2026
Head of Technical Affairs and Problem Selection Committee <i>Scholastic Aptitude Test in Mathematics (SATM), conducted by the student association of the Institute of Mathematics and Applications, Bhubaneswar, as part of its annual techno-cultural fest. The event focuses on spreading awareness about higher education in mathematics and organises training camps for selected high school students in Odisha, India.</i> https://sites.google.com/view/satm-ima	2018

SKILLS

<i>Programming</i>	C, C++, PYTHON, JAVASCRIPT, JAVA, VISUAL BASIC
<i>Typesetting</i>	L ^A T _E X
<i>Markup</i>	HTML
<i>Stylesheet</i>	CSS
<i>Office Software</i>	MS WORD, MS EXCEL, MS POWERPOINT, MS ACCESS
<i>Languages</i>	Bengali (mother tongue) English (professional) German (proficient, A2) French (proficient, A1) Hindi (native) Sanskrit (intermediate)